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Deploying CNNs and Vision Transformers on STM32N6 Edge Devices

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- ① The board and the experimental background
- ② Deployment analysis and results
- ③ Summary of results

Deployment challenges

- **Memory Limits:** Embedded devices have strict RAM/Flash constraints.
- **Low Compute Power:** MCUs/NPUs offer far less throughput than GPUs.
- **Real-Time Needs:** Applications often require low-latency inference.
- **Energy Constraints:** Battery-based systems need efficient models.
- **Model Support:** Transformer deployment (e.g., ViT) is still limited.

Target Board Characteristics

Feature	Description
Processor	Arm Cortex-M55 MCU
Accelerators	Neural-ART, H.264 encoder
GPU	NeoChrom 2.5D GPU
CLK frequency	800 MHz
Memory	4.2 MB SRAM, Octo-SPI flash
Networking	1-Gbit Ethernet with TSN
Debugging	STLINK-V3EC debugger

Table: STM32N6570-DK Board Specifications

YOLOv1 Deployment Results

Metric	Value
MACCs	62.31 M
Model Size	316.7 KB
Activation Memory	252 KB
NPU Execution Rate	95.5%
Inference Time	1.96 ms
Inferences per Second	510.20
TOPS	0.0636

Table: Deployment metrics for `st_yolo_lc_v1`

YOLOv5n Deployment Results

Metric	Value
MACCs	357.05 M
Model Size	2.55 MB
Activation Memory	310.5 KB
NPU Execution Rate	97.5%
Inference Time	23.38 ms
Inferences per Second	42.77
TOPS	0.0306

Table: Deployment metrics for YOLOv5n

ViT Deployment Results

Metric	Value
MACCs	7.87 Billion
Model Size	1.72 MB
Activation Memory	228.0 KB
NPU Execution Rate	53.3%
Inference Time	256.3 ms
Inferences per Second	3.90
TOPS	0.0614

Table: Deployment metrics for Vision Transformer (ViT)

Key Deployment Metrics

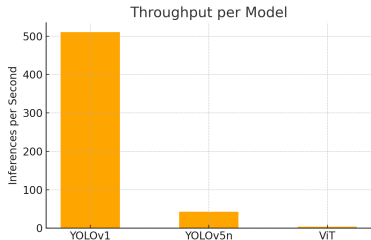
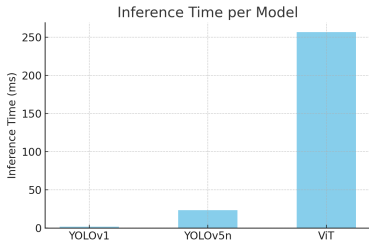


Figure: Inference Time and Throughput for YOLOv1, YOLOv5n, and ViT

- **YOLOv1** is the clear winner for real-time use (tiny, fast, almost fully on NPU).
- **YOLOv5n** trades a bit of speed for better accuracy capacity while still using the NPU widely.
- **ViT** works, but many ops fall back to the M55, so latency explodes.
- Next step: replace Softmax/LayerNorm with NPU-friendly blocks and try a lightweight DETR-style detector.

Q&A